DATA200-LAB1

GitHub link: https://github.com/PINAL99/SJSU_DATA200

Introduction
Object-Oriented Programming (OOP) and Python are used in the development of CheckMyGrade application. This system stores, retrieves, and analyses grade data in order to manage and assess students' academic performance.
The program supports user roles and uses CSV files as database. These roles include student,professor and admin
The system offers functions like searching, sorting, and statistical analysis and provides efficient data handling through file operations.

Classes used :-
User as base class
Student
Professor
Admin
Course
Role

CSV Files :-
System Uses 4 csv files:
studnets.csv
professors.csv
courses.csv
login.csv

Login :-
Reads login.csv
Decrypts stored passwords
Matches credentials

Password Encryption and Decryption
Uses base64 encoding
Stored encrypted password in login.csv
Decrypts during login

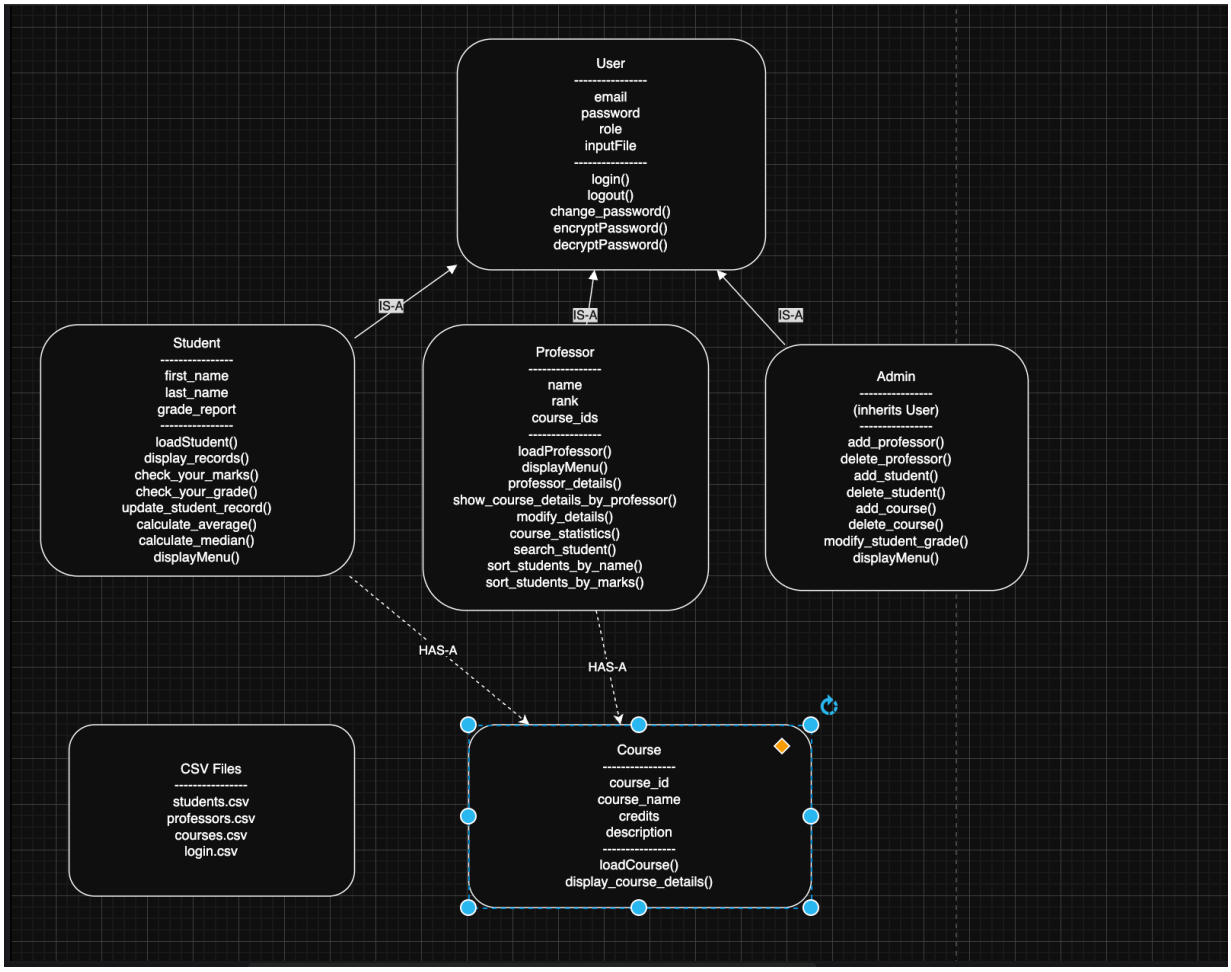
Change Password
Updates password in csv file
Encrypts before saving

Logput
Resets user role to unknown

Main Function :-
User enters email and password
System Validates login
Based on role it open the menu accordingly

IS-A HAS-A Relationship
IS-A
Student is a user
Professor is a user
Admin is a user

HAS-A
Student has a grade report
Professor has a list of course_id
Courses has a course details(id, name, credits)



SCREENSHOTS

Initial screen will ask for email and password
ADMIN/STUDENT/PROFESSOR can login
Change password , logout and exit is common in all three
Change Password will change the password and store it in the encrypted format in login.csv file
Logout will send you to main screen.
Exit will take you out of the code

```
=== Welcome to CheckMyGrade ===
Enter email: admin@abc.com
Enter password: ADMIN001
```

STUDENT
Displays grades and marks and allows updating details

Initial Screen

```
=== Welcome to CheckMyGrade ===
Enter email: sam@abc.com
Enter password: ST12345
Login successful student

----- Student Menu -----
1. View My Records
2. Check My Marks
3. Check My Grades
4. Update My Name
5. Change Password
6. Logout
7. Exit
Enter choice:
```

View My Record
Displays records of the student

```
=== Welcome to CheckMyGrade ===
Enter email: sam@abc.com
Enter password: ST12345
Login successful student

----- Student Menu -----
1. View My Records
2. Check My Marks
3. Check My Grades
4. Update My Name
5. Change Password
6. Logout
7. Exit
Enter choice: 1
Student record:
Name   : Sam Carpenter
Email  : sam@abc.com
Course Details:
Course ID : DATA200
Grade     : C
Marks     : 66
```

Check My marks
Shows the marks of the student

```
----- Student Menu -----
1. View My Records
2. Check My Marks
3. Check My Grades
4. Update My Name
5. Change Password
6. Logout
7. Exit
Enter choice: 2
Here are your marks
DATA200 : 66
```

Check My Grades
Shows the grade of the student

```
----- Student Menu -----
1. View My Records
2. Check My Marks
3. Check My Grades
4. Update My Name
5. Change Password
6. Logout
7. Exit
Enter choice: 3
Here are your grades
DATA200 : C
```

Update Name
Updates Student records as student has the right to only change names and not marks or grades.
Initial

students

email	firstName	lastName	courseId	grade	marks
sam@abc.com	Sam	Carpenter	DATA200	A	96
jane@abc.com	Jane	Doe	DATA200	B	85
alex@abc.com	Alex	Smith	CS101	A	91

```
----- Student Menu -----
1. View My Records
2. Check My Marks
3. Check My Grades
4. Update My Name
5. Change Password
6. Logout
7. Exit
Enter choice: 4
Enter new first name: Samel
Enter new last name: Carton
Student details updated successfully
```

Updated Name

students

email	firstName	lastName	courseId	grade	marks
sam@abc.com	Sam	Carpenter	DATA200	A	96
jane@abc.com	Jane	Doe	DATA200	B	85
alex@abc.com	Alex	Smith	CS101	A	91

Change Password:
Initial Password Encrypted format (Decrypted :ST12345)

login

email	password	role
micheal@abc.com	QVExMDEzNA==	professor
sam@abc.com	U1QxMjM0NQ==	student
admin@abc.com	QURNSU4wMDE=	admin

Changed

```
=== Welcome to CheckMyGrade ===
Enter email: sam@abc.com
Enter password: ST12345
Login successful student

----- Student Menu -----
1. View My Records
2. Check My Marks
3. Check My Grades
4. Update My Name
5. Change Password
6. Logout
7. Exit
Enter choice: 5
Enter new password: 12345
Password updated successfully
```

Updated Encrypted

login

email	password	role
micheal@abc.com	QVExMDEzNA==	professor
sam@abc.com	MTIzNDU=	student
admin@abc.com	QURNSU4wMDE=	admin

Login using New Password (12345)

```
----- Student Menu -----
1. View My Records
2. Check My Marks
3. Check My Grades
4. Update My Name
5. Change Password
6. Logout
7. Exit
Enter choice: 6
Logged out successfully

=== Welcome to CheckMyGrade ===
Enter email: sam@abc.com
Enter password: 12345
Login successful student

----- Student Menu -----
1. View My Records
2. Check My Marks
3. Check My Grades
4. Update My Name
5. Change Password
6. Logout
7. Exit
Enter choice:
```

Handled Error like invalid Choice

```
----- Student Menu -----
1. View My Records
2. Check My Marks
3. Check My Grades
4. Update My Name
5. Change Password
6. Logout
7. Exit
Enter choice: 9
Error - Invalid choice
```

ADMIN
Full control over system and can manage all data

Email : admin@abc.com
Password : ADMIN001

```
=== Welcome to CheckMyGrade ===
Enter email: admin@abc.com
Enter password: ADMIN001
Login successful admin

----- Admin Menu -----
1. Add Professor
2. Delete Professor
3. Add Student
4. Delete Student
5. Add Course
6. Delete Course
7. Modify Student Grade
8. Change Password
9. Logout
10. Exit
Enter choice:
```

Add Professor :
Added A Professor named "Ken" to professor.csv

```
=== Welcome to CheckMyGrade ===
Enter email: admin@abc.com
Enter password: ADMIN001
Login successful admin

----- Admin Menu -----
1. Add Professor
2. Delete Professor
3. Add Student
4. Delete Student
5. Add Course
6. Delete Course
7. Modify Student Grade
8. Change Password
9. Logout
10. Exit
Enter choice: 1
Enter professor email: ken@abc.com
Enter name: Ken
Enter rank: Professor
Enter course id: DATA209
Professor added successfully
```

professors

email	name	rank	courseId
micheal@abc.com	Micheal John	Senior Professor	DATA200
susan@abc.com	Susan Lee	Associate Professor	CS101
david@abc.com	David Kim	Assistant Professor	MATH150
ken@abc.com	Ken	Professor	DATA209

Delete Professor :
Deletes Prof Ken. From professor.csv

```
----- Admin Menu -----
1. Add Professor
2. Delete Professor
3. Add Student
4. Delete Student
5. Add Course
6. Delete Course
7. Modify Student Grade
8. Change Password
9. Logout
10. Exit
Enter choice: 2
Enter professor email to delete: ken@abc.com
Professor deleted successfully
```

professors

email	name	rank	courseId
micheal@abc.com	Micheal John	Senior Professor	DATA200
susan@abc.com	Susan Lee	Associate Professor	CS101
david@abc.com	David Kim	Assistant Professor	MATH150

Add Student :
Adds student along with all its information as well as Grades and marks

```
----- Admin Menu -----
1. Add Professor
2. Delete Professor
3. Add Student
4. Delete Student
5. Add Course
6. Delete Course
7. Modify Student Grade
8. Change Password
9. Logout
10. Exit
Enter choice: 3
Enter student email: jay@abc.com
Enter first name: Jay
Enter last name: Han
Enter course id: DATA200
Enter grade: B
Enter marks: 87
Student added successfully
```

students

email	firstName	lastName	courseId	grade	marks
sam@abc.com	Sam	Carpenter	DATA200	A	96
jane@abc.com	Jane	Doe	DATA200	B	85
alex@abc.com	Alex	Smith	CS101	A	91
jay@abc.com	Jay	Han	DATA200	B	87

Delete Student:
Removes student along with all its information as well as Grades and marks

```
----- Admin Menu -----
1. Add Professor
2. Delete Professor
3. Add Student
4. Delete Student
5. Add Course
6. Delete Course
7. Modify Student Grade
8. Change Password
9. Logout
10. Exit
Enter choice: 4
Enter student email to delete: jay@abc.com
Student deleted successfully
```

students

email	firstName	lastName	courseId	grade	marks
sam@abc.com	Sam	Carpenter	DATA200	A	96
jane@abc.com	Jane	Doe	DATA200	B	85
alex@abc.com	Alex	Smith	CS101	A	91

Add Course
Adds the course to course.csv

```
----- Admin Menu -----
1. Add Professor
2. Delete Professor
3. Add Student
4. Delete Student
5. Add Course
6. Delete Course
7. Modify Student Grade
8. Change Password
9. Logout
10. Exit
Enter choice: 5
Enter course ID: DATA320
Enter course name: GenAI
Enter credits: 4
Enter course description: AI
Course added successfully
```

courses

id	name	credits	description
DATA200	Data Science	3	Provides insight about DS and Python
CS101	Intro to Computer Science	3	Basics of programming and algorithms
MATH150	Calculus I	3	Introduction to differential calculus
DATA320	GenAI	4	AI

Delete Course
Removes the course from course.csv

```
----- Admin Menu -----
1. Add Professor
2. Delete Professor
3. Add Student
4. Delete Student
5. Add Course
6. Delete Course
7. Modify Student Grade
8. Change Password
9. Logout
10. Exit
Enter choice: 6
Enter course ID to delete: DATA320
Course deleted successfully
```

courses

id	name	credits	description
DATA200	Data Science	3	Provides insight about DS and Python
CS101	Intro to Computer Science	3	Basics of programming and algorithms
MATH150	Calculus I	3	Introduction to differential calculus

Modify Student Grade
Changes grades for students
Here it changes Sam's grade from A to C

students

email	firstName	lastName	courseId	grade	marks
sam@abc.com	Sam	Carpenter	DATA200	A	96
jane@abc.com	Jane	Doe	DATA200	B	85
alex@abc.com	Alex	Smith	CS101	A	91

```
----- Admin Menu -----
1. Add Professor
2. Delete Professor
3. Add Student
4. Delete Student
5. Add Course
6. Delete Course
7. Modify Student Grade
8. Change Password
9. Logout
10. Exit
Enter choice: 7
Enter student email: sam@abc.com
Enter course ID: DATA200
Enter new grade: C
Enter new marks: 66
Student sam@abc.com's grade updated successfully for course DATA200
```

students

email	firstName	lastName	courseId	grade	marks
sam@abc.com	Sam	Carpenter	DATA200	C	66
jane@abc.com	Jane	Doe	DATA200	B	85
alex@abc.com	Alex	Smith	CS101	A	91

PROFESSOR
Shows course statistics and allows sorting and searching with time

Email:micheal.abc.com
Password : AQ10134

Initial Screen

```
=== Welcome to CheckMyGrade ===
Enter email: micheal@abc.com
Enter password: AQ10134
Login successful professor

----- Professor Menu -----
1. View My Details
2. Show Course Details
3. Modify My Details
4. Course Statistics
5. Search Student
6. Sort Students by Name
7. Sort Students by Marks
8. Change Password
9. Logout
10. Exit
Enter choice:
```

View My detail
Shows professor details

```
----- Professor Menu -----
1. View My Details
2. Show Course Details
3. Modify My Details
4. Course Statistics
5. Search Student
6. Sort Students by Name
7. Sort Students by Marks
8. Change Password
9. Logout
10. Exit
Enter choice: 1
--------------------------------------------------
Email      : micheal@abc.com
Name       : Micheal John
Rank       : Senior Professor
Courses    : DATA200
--------------------------------------------------
```

Show Course Details
Shows the his course details

```
----- Professor Menu -----
1. View My Details
2. Show Course Details
3. Modify My Details
4. Course Statistics
5. Search Student
6. Sort Students by Name
7. Sort Students by Marks
8. Change Password
9. Logout
10. Exit
Enter choice: 2
Course details:
Course ID      : DATA200
Course Name    : Data Science
Credits        : 3
Description    : Provides insight about DS and Python
```

Modify Details
Changes his Details like from senior Professor to Professor in this case

Initially

professors

email	name	rank	courseId
micheal@abc.com	Micheal John	Senior Professor	DATA200
susan@abc.com	Susan Lee	Associate Professor	CS101
david@abc.com	David Kim	Assistant Professor	MATH150

```
----- Professor Menu -----
1. View My Details
2. Show Course Details
3. Modify My Details
4. Course Statistics
5. Search Student
6. Sort Students by Name
7. Sort Students by Marks
8. Change Password
9. Logout
10. Exit
Enter choice: 3
Enter new name: Michael
Enter new rank: Professor
Professor details updated successfully
```

professors

email	name	rank	courseId
micheal@abc.com	Michael	Professor	DATA200
susan@abc.com	Susan Lee	Associate Professor	CS101
david@abc.com	David Kim	Assistant Professor	MATH150

Course Statistics
Shows the MEAN and MEDIAN for his course as a performance metric

```
----- Professor Menu -----
1. View My Details
2. Show Course Details
3. Modify My Details
4. Course Statistics
5. Search Student
6. Sort Students by Name
7. Sort Students by Marks
8. Change Password
9. Logout
10. Exit
Enter choice: 4

Statistics for course DATA200:
Average marks: 75.5
Median marks : 75.5
```

Search Student
Searches the student in his course as well as showcases the time taken for it to search
In this case it took).003523 seconds to search a student named Jane

```
----- Professor Menu -----
1. View My Details
2. Show Course Details
3. Modify My Details
4. Course Statistics
5. Search Student
6. Sort Students by Name
7. Sort Students by Marks
8. Change Password
9. Logout
10. Exit
Enter choice: 5
Enter student email to search: jane@abc.com
Found student: Jane Doe - jane@abc.com
Time taken: 0.003523 seconds
```

Sort student by Name
Places students in ascending order via their name

```
----- Professor Menu -----
1. View My Details
2. Show Course Details
3. Modify My Details
4. Course Statistics
5. Search Student
6. Sort Students by Name
7. Sort Students by Marks
8. Change Password
9. Logout
10. Exit
Enter choice: 6
Students sorted by name:
Jane Doe - jane@abc.com - Course: DATA200
Samel Carton - sam@abc.com - Course: DATA200
```

Sort Student by Marks
Places students in ascending order via their marks

```
----- Professor Menu -----
1. View My Details
2. Show Course Details
3. Modify My Details
4. Course Statistics
5. Search Student
6. Sort Students by Name
7. Sort Students by Marks
8. Change Password
9. Logout
10. Exit
Enter choice: 7
Enter course ID to sort students by marks: DATA200
Students sorted by marks for course DATA200:
Jane Doe - 85
Samel Carton - 66
```

Handles Wrong Password and Wrong Email

```
=== Welcome to CheckMyGrade ===
Enter email: abc@abc.com
Enter password: 1234
Error - Email not found

=== Welcome to CheckMyGrade ===
Enter email: sam@abc.com
Enter password: 1234
Error - Incorrect password
```

UNIT TESTING :

Tests covered

```python
        with open(self.students_csv, "w", newline="") as f:
            writer = csv.DictWriter(f, fieldnames=["email","firs
            writer.writeheader()
            writer.writerows(students)
        with open(self.students_csv, "r") as f:
            students_loaded = list(csv.DictReader(f))
        start_sort = time.time()
        students_sorted_marks = sorted(students_loaded, key=lamb
        print(f"Sorting by marks time: {time.time()-start_sort:.
        start_sort = time.time()
        students_sorted_email = sorted(students_loaded, key=lamb
        print(f"Sorting by email time: {time.time()-start_sort:.
        self.assertTrue(int(students_sorted_marks[0]["marks"]) <
        self.assertTrue(students_sorted_email[0]["email"] <= stu
        print(f">>> Time taken by test_student_sorting: {time.ti

if __name__ == "__main__":
    unittest.main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(base) atharvamundada@Atharvas-MacBook-Air Downloads % /usr/bin/python3 /
>>> Time taken by test_course_add: 0.000 seconds
.>>> Time taken by test_course_delete: 0.000 seconds
.>>> Time taken by test_course_modify: 0.000 seconds
.>>> Time taken by test_professor_add: 0.000 seconds
.>>> Time taken by test_professor_delete: 0.000 seconds
.>>> Time taken by test_professor_modify: 0.000 seconds
.>>> Time taken by test_student_add: 0.004 seconds
.>>> Time taken by test_student_delete: 0.001 seconds
.>>> Time taken by test_student_modify: 0.000 seconds
.Search time: 0.000148 seconds
>>> Time taken by test_student_search: 0.000 seconds
.Sorting by marks time: 0.000004 seconds
Sorting by email time: 0.000001 seconds
>>> Time taken by test_student_sorting: 0.000 seconds
.
----------------------------------------------------------------------
Ran 11 tests in 0.012s

OK
```

CONCLUSION:

The system offers functions like searching, sorting, and statistical analysis and provides efficient data handling through file operations.

The CheckMyGrade program illustrates:
Effective application of object-oriented design,
file-based data management,
real-world academic system implementation,
combining search optimisation, statistics, and security

The system can be expanded into a complete GUI or online application and is scalable.